

Ex.No: 5	Implement the Digital Signature Algorithm (DSA).
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AIM:

ALGORITHM:

CODE:

```
import hashlib

def modular_inverse(number, modulus): return

    pow(number, -1, modulus)

def message_hash(message, q):

    hash_value = hashlib.sha256(message.encode("utf-8")).digest()

    hash_integer = int.from_bytes(hash_value, byteorder="big") return

    hash_integer % q

def generate_signature(message, p, q, g, private_key, k):

    hash_value = message_hash(message, q)

    r = pow(g, k, p) % q

    k_inverse = modular_inverse(k, q)

    s = (k_inverse * (hash_value + private_key * r)) % q return r,

    s, hash_value

def verify_signature(message, signature, p, q, g, public_key): r, s

    = signature

    if not (0 < r < q and 0 < s < q): return

        False, None, None, None

    hash_value = message_hash(message, q) w

    = modular_inverse(s, q)

    u1 = (hash_value * w) % q u2 =

    (r * w) % q

    v = ((pow(g, u1, p) * pow(public_key, u2, p)) % p) % q

    return v == r, v, u1, u2

def main():

    p = 23

    q = 11
```

```

g = 4
private_key = 3
public_key = pow(g, private_key, p) k =
7
message = input("Enter the message: ")
signature_r, signature_s, hash_value = generate_signature( message,
    p, q, g, private_key, k
)
print("\n----- DSA Parameters   ")
print("Prime Number (p) :", p)
print("Prime Divisor (q) :", q)
print("Generator (g) :", g)
print("\n----- Key Generation      ")
print("Private Key (x) :", private_key)
print("Public Key (y) :", public_key)
print("\n----- Signature Generation    ")
print("Message :", message) print("Message
Hash :", hash_value) print("Random
Number (k) :", k) print("Signature r :",
signature_r) print("Signature s :",
signature_s)
print("Digital Signature :", (signature_r, signature_s)) valid,
v, u1, u2 = verify_signature(
    message,
    (signature_r, signature_s), p,
    q, g, public_key
)

```

```
print("\n----- Signature
```

```
Verification
```

```
print("u1 :", u1)
```

```
print("u2 :", u2)
```

```
print("v :", v)
```

```
print("r,signat
```

```
ure_r) if valid:
```

```
    print("\nSignature Status : VALID") else:
```

```
    print("\nSignature Status : INVALID") if
```

```
name_____==" main
```

```
    ": main()
```

```
]
```

```
jj
```

```
Enter the message: HELLO
```

```
----- DSA Parameters -----
```

```
Prime Number (p) : 23
```

```
Prime Divisor (q) : 11
```

```
Generator (g) : 4
```

```
----- Key Generation -----
```

```
Private Key (x) : 3
```

```
Public Key (y) : 18
```

```
----- Signature Generation -----
```

```
Message : HELLO
```

```
Message Hash : 7
```

```
Random Number (k) : 7
```

```
Signature r : 8
```

```
Signature s : 6
```

```
Digital Signature : (8, 6)
```

```
----- Signature Verification -----
```

```
u1 : 3
```

```
u2 : 5
```

```
v : 8
```

```
r : 8
```

```
Signature Status : VALID
```

RESULT: